

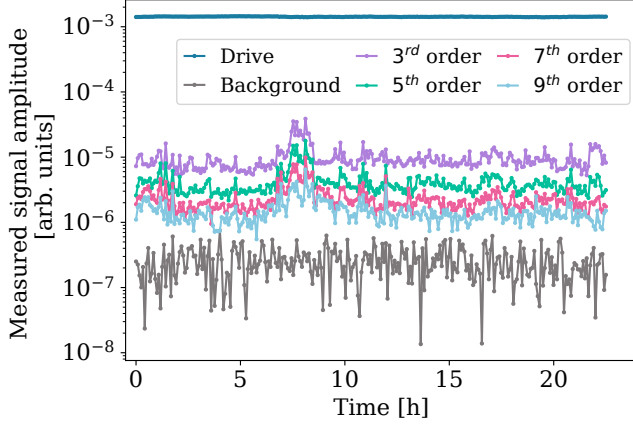


FIG. 3. IMPs measured repeatedly for 25h, in 5-minute intervals, at a constant drive power level of $\sim 10^3$ photons.

TABLE I. Power dependence of the measured IMPs in Fig. 2d), and of the modeled IMPs in Fig. 4, fitted to $\langle n \rangle^k \cdot 10^l$.

IMP order	Experimental fit k	Model fit k
3	0.41 ± 0.05	0.42
5	0.38 ± 0.04	0.43
7	0.42 ± 0.04	0.44
9	0.48 ± 0.05	0.45

IV. NONLINEAR PARAMETER RECONSTRUCTION

In this section, we detail the modelling and reconstruction of the nonlinear loss due to parasitic TLSs. Our reconstruction algorithm is based on fitting the linear and nonlinear oscillator parameters to a single intermodulation spectrum by harmonic balance [33, 34].

We model our resonator as a driven harmonic oscillator with non-linear damping caused by parasitic TLSs. The Heisenberg-Langevin equation of motion describing the dynamics of the resonator in the rotating-wave approximation is then

$$\dot{\hat{a}} = -i\omega_0 \hat{a} - \frac{\kappa_0 + \kappa_{ext}}{2} \hat{a} - \frac{\kappa_{TLS}(|a|)}{2} \hat{a} - \sqrt{\kappa_{ext}} a_{in}, \quad (3)$$

where $\hat{a}$ and a_{in} are the time-dependent intra-cavity modes and the input drives respectively, ω_0 the resonant frequency, κ_0 the internal linear loss rate, κ_{ext} the external coupling to the transmission line, and $\kappa_{TLS}(|a|)$ the power-dependent loss due to parasitic TLSs. We consider a power-dependent damping of the standard form of Eq. (2), giving

$$\kappa_{TLS}(|a|)a = \frac{\kappa_{TLS}}{[1 + (|a|/a_c)^2]^\beta}, \quad (4)$$

where $\kappa_{TLS} = f_r F \delta_{TLS}^0$ represents the TLS loss rate, and $a_c^2 = n_c$ the critical photon number.

We drive the system with two tones, separated by Δ , such that the IMPs created are within the resonator's linewidth, where the sensitivity is enhanced. When the drive frequencies are periodic in some measurement time $T = 2\pi/\Delta$, and the nonlinearities and drives are sufficiently weak, the system's response is a discrete spectrum at frequencies that correspond to integer multiples of Δ .

We first solve the forward problem by numerically integrating Eq. (3) using `scipy.integrate.ode` from the SciPy library [35] to obtain the time-dependent intra-cavity field. The output field is then obtained using the input-output relation $\hat{a}_{out} = \sqrt{\kappa_{ext}} \hat{a} + a_{in}$. Fig. 4) shows the four strongest IMPs as a function of drive power simulated with experimentally relevant parameters. For strong drives $\langle n \rangle \gg 1$, the IMP power shows a power-law scaling proportional to $\langle n \rangle^k$, with parameters k laid out in Table I. This behaviour matches the experimental results shown in Fig. 2 b). We also simulate a sweep of the drive tone frequency across the resonance and analyze the power dependence of the IMPs as a function of β (see Appendix E for details).

To reconstruct the nonlinear parameters of the system, we expand the TLS damping in Eq. 4 in a power series, transforming the equation of motion to

$$\dot{\hat{a}} = -i\omega_0 \hat{a} - \sum_{n=1}^N c_n |a|^{n-1} \hat{a} - \sqrt{\kappa_{ext}} a_{in}, \quad (5)$$

which is linear in the coefficients c_n . The coefficients are independent of the proposed model, allowing the application of the reconstruction algorithm to various types of nonlinearities [17, 36], including unknown ones.

In the experiment we measure only a limited number of frequencies with good SNR, corresponding to the two drive tones and 14 IMPs. These frequencies, denoted as ω_k with Fourier component $\hat{A}_k$, represent a partial spectral response of the system. Specifically, $\hat{A}_k$ corresponds to the intracavity field amplitude at the frequency ω_k .

By Fourier transforming and rewriting Eq. (5) in matrix form, we arrive at an expression linear in the coefficients c_n [17]

$$\sum_l H_{kl} p_l = -A_{in,k}, \quad (6)$$

where the k th row of the matrix H_{kl} is given by

$$H_k = \left(i\omega_k \hat{A}_k \quad i\hat{A}_k \quad \hat{A}_k \quad \mathcal{F}\{|a|^2 \hat{a}\}_k \quad \dots \quad \mathcal{F}\{|a|^{N-1} \hat{a}\}_k \right),$$

and p_l are the components of a vector containing the unknown parameters

$$\sqrt{\kappa_{ext}} p^T = (1 \quad \omega_0 \quad c_1 \quad \dots \quad c_N).$$

We solve Eq. (6) by numerical pseudo-inverse of the matrix H to find the best-fit vector p , which includes the polynomial coefficients c_n . To extract the TLS damping parameters κ_{TLS} , a_c and β from the polynomial expansion coefficients and to compare our harmonic balance results with the circle fit analysis, we perform a